

Spin Precession

- 1.) Historical note: What were Otto Stern and Walther Gerlach looking for when they did their famous experiment in 1922? Hint: spin was not even hypothesized until 1925 and wasn't fully worked out until 1927 (by Pauli).

Solution: Spatial quantization of orbitals. They were working with Bohr's model of the atom, and hypothesizing that his quantized, non-radiating orbitals (still thought to be circular planetary-like orbits of moving charge) should have a small orbital magnetic moment that could lead to a deflection in a small inhomogeneous magnetic field. They got lucky.

- 2.) Consider the operator:

$$\hat{S}_{\hat{n}} = \hat{S} \cdot \hat{n} = \hat{S} \cdot (\sin(\theta) \cos(\phi), \sin(\theta) \sin(\phi), \cos(\theta))$$

Where $\hat{S} = \hbar \hat{\sigma}/2$ and $\hat{\sigma}$ is a vector of the Pauli matrices.

- (a) Give a physical interpretation of this operator.
(b) What are this operator's eigenvalues?

Solution:

(a) Solution: This is the spin operator along some direction $\hat{n}$, which is at the angles θ, ϕ .

(b) Starting from $\hat{S}_{\hat{n}} = \hat{S} \cdot \hat{n} = \hat{S} \cdot (\sin(\theta) \cos(\phi), \sin(\theta) \sin(\phi), \cos(\theta))$ we can write this as a matrix:

$$\begin{aligned} \hat{S}_{\hat{n}} &= \frac{\hbar}{2} \begin{pmatrix} \cos(\theta) & \sin(\theta) \cos(\phi) - i \sin(\theta) \sin(\phi) \\ \sin(\theta) \cos(\phi) + i \sin(\theta) \sin(\phi) & -\cos(\theta) \end{pmatrix} \\ &= \frac{\hbar}{2} \begin{pmatrix} \cos(\theta) & \sin(\theta) e^{-i \sin(\phi)} \\ \sin(\theta) e^{i \sin(\phi)} & -\cos(\theta) \end{pmatrix} \end{aligned}$$

So, the eigenvalues are given by

$$\left| \begin{pmatrix} \frac{\hbar}{2} \cos(\theta) - \lambda & \frac{\hbar}{2} \sin(\theta) e^{-i\phi} \\ \frac{\hbar}{2} \sin(\theta) e^{i\phi} & -\frac{\hbar}{2} \cos(\theta) - \lambda \end{pmatrix} \right| = -\left(\frac{\hbar}{2}\right)^2 \cos^2(\theta) + \lambda^2 - \left(\frac{\hbar}{2}\right)^2 \sin^2(\theta) = 0 \Rightarrow \lambda = \pm \frac{\hbar}{2}$$

This fits our expectation, since this is just a spin operator of the same size/spin as our familiar old $\hat{S}_z$, just oriented along a new axis.

- 3.) In trying to find the eigenvectors of $\sigma^{\rightarrow} \cdot \hat{n}$, your friend keeps getting $|n+\rangle = \begin{pmatrix} \cos(\theta/2) e^{-i\phi} \\ \sin(\theta/2) \end{pmatrix}$ and you keep getting $|n+\rangle = \begin{pmatrix} \cos(\theta/2) e^{-i\phi/2} \\ \sin(\theta/2) e^{i\phi/2} \end{pmatrix}$. What, if anything, are you (or your friend) doing wrong?

Solution: Nothing! It is just an overall phase! No problem.

4.) Consider the state $\frac{1}{\sqrt{13}} \begin{pmatrix} 2 \\ 3i \end{pmatrix}$.

- (a) What is the physical interpretation of this state? What are the possible outcomes and corresponding probabilities for making $\hat{S}_z$ measurements?
- (b) Explain how one can produce such a state in the lab. HINT: what equipment / setup would you need and how would you orient it?

You can use that the eigenstates of

$$\hat{S}_{\hat{n}} = \hat{\vec{S}} \cdot \hat{n} = \hat{\vec{S}} \cdot (\sin(\theta) \cos(\phi), \sin(\theta) \sin(\phi), \cos(\theta))$$

are (in the S_z basis)

$$|+n\rangle = \cos(\theta/2) |\uparrow\rangle + \sin(\theta/2) e^{i\phi} |\downarrow\rangle$$

$$|-n\rangle = \sin(\theta/2) |\uparrow\rangle - \cos(\theta/2) e^{i\phi} |\downarrow\rangle$$

Solution:

(a) This is a state that has a well-defined spin projection along SOME axis, but not along x, y, or z. The probability of a measurement of $\hat{S}_z$ yielding $+\hbar/2$ is 4/13, and of yielding $-\hbar/2$ is 9/13.

(b) As part of homework, students are finding the eigenstates to the $\hat{S}_{\hat{n}} = \hat{\vec{S}} \cdot \hat{n} = \hat{\vec{S}} \cdot (\sin(\theta) \cos(\phi), \sin(\theta) \sin(\phi), \cos(\theta))$ matrix. They should find,

$$|+n\rangle = \cos(\theta/2) |\uparrow\rangle + \sin(\theta/2) e^{i\phi} |\downarrow\rangle$$

$$|-n\rangle = \sin(\theta/2) |\uparrow\rangle - \cos(\theta/2) e^{i\phi} |\downarrow\rangle$$

If we select $|+n\rangle$ from a magnet oriented along the $+n$ direction, we need the coefficient of $|\downarrow\rangle$ to be purely imaginary. So, we set $\phi = \pi/2$. The ratio of the coefficients of the two orthogonal basis states is $\frac{\sin(\theta/2)}{\cos(\theta/2)} = \tan(\theta/2) = \frac{3}{2} \Rightarrow \theta = 2 \tan^{-1}(3/2) = 113^\circ$.

5.) Suppose you prepare a state $|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$ at time $t = 0$.

- (a) Immediately after this state is prepared (after a time much smaller than characteristic time scales for the system), find the probability that a measurement of spin projection onto the z axis yields $+\hbar/2$.
- (b) Now let us switch on a magnetic field $\vec{B} = B_0 \hat{i}$. Write down the time-dependent state.
- (c) Find the probability that a measurement of the spin projection on the z axis will yield $+\hbar/2$, as a function of time. Hint: you may use the fact that $\hat{H} = -\vec{\mu} \cdot \vec{B} = \left(\frac{eg}{2m}\right) \hat{\vec{S}} \cdot \vec{B}$, where $g_e \approx 2$, and note (cryptically) the units of $\left(\frac{eB_0}{m}\right)$.

Solution:

$$(a) P_{+z} = \left| \langle + | \psi \rangle \right|^2 = \left| (1, 0) \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \right|^2 = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} = 50\%$$

(b) We will first do this in the basis that diagonalizes the Hamiltonian:

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\psi\rangle = e^{-i\omega \hat{S}_x t/\hbar} |\psi\rangle = e^{-i\omega \hat{S}_x t/\hbar} \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} (|+\rangle_x + |-\rangle_x) + \frac{i}{\sqrt{2}} (|+\rangle_x - |-\rangle_x) \right)$$

$$|\psi(t)\rangle = \frac{1}{2} \left((e^{-i\omega t/2} |+\rangle_x + e^{i\omega t/2} |-\rangle_x) + i(e^{-i\omega t/2} |+\rangle_x - e^{i\omega t/2} |-\rangle_x) \right)$$

This is a fine form for the time-dependent state. But since we know we eventually want to take the inner product with eigenstates of $\hat{S}_z$, we will go ahead and express this in terms of eigenstates of $\hat{S}_z$:

$$|\psi(t)\rangle = \frac{1}{2\sqrt{2}} \left((e^{-i\omega t/2} (|+\rangle_z + |-\rangle_z) + e^{i\omega t/2} (|+\rangle_z - |-\rangle_z)) + i(e^{-i\omega t/2} (|+\rangle_z + |-\rangle_z) - e^{i\omega t/2} (|+\rangle_z - |-\rangle_z)) \right)$$

(c)

$$\begin{aligned} P_{+z} &= \left| \langle + | \psi \rangle \right|^2 \\ &= \left| \langle + | \frac{1}{2\sqrt{2}} \left((e^{-i\omega t/2} (|+\rangle_z + |-\rangle_z) + e^{i\omega t/2} (|+\rangle_z - |-\rangle_z)) + i(e^{-i\omega t/2} (|+\rangle_z + |-\rangle_z) - e^{i\omega t/2} (|+\rangle_z - |-\rangle_z)) \right) \right|^2 \\ &= \frac{1}{8} \left| \left((e^{-i\omega t/2} + e^{i\omega t/2}) + i(e^{-i\omega t/2} - e^{i\omega t/2}) \right) \right|^2 \\ &= \frac{1}{8} \left((e^{-i\omega t/2} + e^{i\omega t/2}) + i(e^{-i\omega t/2} - e^{i\omega t/2}) \right) \left((e^{i\omega t/2} + e^{-i\omega t/2}) - i(e^{i\omega t/2} - e^{-i\omega t/2}) \right) \\ &= \frac{1}{8} \left((2 + e^{-i\omega t} + e^{i\omega t}) + (2 - e^{-i\omega t} - e^{i\omega t}) - i(e^{i\omega t} - e^{-i\omega t}) + i(e^{-i\omega t} - e^{i\omega t}) \right) \\ &= \frac{1}{8} \left(4 + \frac{2}{i} (e^{i\omega t} - e^{-i\omega t}) \right) = \frac{1}{2} \left(1 + \frac{1}{2i} (e^{i\omega t} - e^{-i\omega t}) \right) \\ &= \frac{1}{2} (1 + \sin(\omega t)) \end{aligned}$$

Note that at $t = 0$, $P \rightarrow \frac{1}{2}$, which makes sense, since the original state was an eigenstate of $\hat{S}_y$.

6.) Suppose a particle is prepared in an eigenstate of $\hat{S}_z$ at time zero: $|\psi(t=0)\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. But right at time

zero, a magnetic field is switched on with $\vec{B} = B_0 \hat{z}$.

(a) Use intuition to argue whether any of the quantities $\langle \hat{S}_x(t) \rangle$, $\langle \hat{S}_y(t) \rangle$, or $\langle \hat{S}_z(t) \rangle$ change in time.

(b) Write out the time-dependent state.

(c) Check your answer to (a) with explicit calculation.

Solution:

(a) One can consider the precessing vector model to see that there can be no non-trivial time evolution of the expectation value of $\langle \hat{S}_x(t) \rangle$: the spin vector (or rather two constituent vectors making it up) precess around the x axis. Their projections along the x axis cannot change in time. The same argument cannot be made for other axes. It seems likely that $\langle \hat{S}_z(t) \rangle$ will change in time, since it starts out at a maximum and the vectors precess around a different axis. However, since the initial expectation value of $\langle \hat{S}_y(t) \rangle$ is zero, it is not obvious whether this will change in time or whether the y components of the vectors will always cancel.

(b) The Hamiltonian for a magnetic moment in a magnetic field is $\hat{H} = -\vec{\mu} \cdot \vec{B} = \left(\frac{eg}{2m}\right) \hat{S} \cdot \vec{B} \rightarrow \left(\frac{e}{m}\right) \hat{S} \cdot \vec{B} \rightarrow \left(\frac{eB_0}{m}\right) \hat{S}_x \equiv \omega \hat{S}_x$.

To write the time-dependent state, we act on the initial state with the time-evolution operator:

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar} |\psi(t=0)\rangle = e^{-i\omega t \hat{S}_x/\hbar} |\psi(t=0)\rangle$$

Because the initial state is not an eigenstate of $\hat{S}_x(t)$, we cannot insert eigenvalues for the operator, unless we write the initial state as a superposition of $\hat{S}_x(t)$ eigenstates. That is $|\psi(t=0)\rangle = |\uparrow\rangle_z = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right) = \frac{1}{\sqrt{2}} (|+\rangle_x + |-\rangle_x)$.

Thus, we can write:

$$|\psi(t)\rangle = e^{-i\omega t \hat{S}_x/\hbar} |\psi(t=0)\rangle = \frac{1}{\sqrt{2}} (e^{-i\omega t/2} |+\rangle_x + e^{i\omega t/2} |-\rangle_x) = \begin{pmatrix} \cos(\omega t/2) \\ -i \sin(\omega t/2) \end{pmatrix}.$$

(c) Given the spinor from (b):

$$\begin{aligned} \langle \hat{S}_x(t) \rangle &= \langle \psi(t) | \hat{S}_x | \psi(t) \rangle = (\cos(\omega t/2), i \sin(\omega t/2)) \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \cos(\omega t/2) \\ -i \sin(\omega t/2) \end{pmatrix} \\ &= (\hbar/2)(\cos(\omega t/2), i \sin(\omega t/2)) \begin{pmatrix} -i \sin(\omega t/2) \\ \cos(\omega t/2) \end{pmatrix} = 0 \end{aligned}$$

As we reasoned in part (a).

$$\begin{aligned} \langle \hat{S}_y(t) \rangle &= \langle \psi(t) | \hat{S}_y | \psi(t) \rangle = (\cos(\omega t/2), i \sin(\omega t/2)) \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} \cos(\omega t/2) \\ -i \sin(\omega t/2) \end{pmatrix} \\ &= (\hbar/2)(\cos(\omega t/2), i \sin(\omega t/2)) \begin{pmatrix} -\sin(\omega t/2) \\ i \cos(\omega t/2) \end{pmatrix} = 0 \end{aligned}$$

We did not know for sure that this would be the case, but we suspected it could be. Finally,

$$\begin{aligned} \langle \hat{S}_z(t) \rangle &= \langle \psi(t) | \hat{S}_z | \psi(t) \rangle = (\cos(\omega t/2), i \sin(\omega t/2)) \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos(\omega t/2) \\ -i \sin(\omega t/2) \end{pmatrix} \\ &= (\hbar/2)(\cos(\omega t/2), i \sin(\omega t/2)) \begin{pmatrix} \cos(\omega t/2) \\ i \sin(\omega t/2) \end{pmatrix} = (\hbar/2)(\cos^2(\omega t/2) - \sin^2(\omega t/2)) = (\hbar/2) \cos(\omega t) \end{aligned}$$

We knew there had to be non-trivial time dependence for this. Note also that the boundary condition matches, in that the state is spin-up at time zero.

7.) Two friends are preparing spin states.

Friend A is in one room and prepares a beam of particles in the state $|S_x; +\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Friend B is in a different room and prepares a different beam in a 50-50 mixture of the two states:

$$|S_z; +\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } |S_z; -\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Friends A and B disagree about whether these two beams are the same thing. One says that those cases are different, both experimentally and mathematically, and the other says they are the same, both experimentally and mathematically. What do you think? Can you offer the incorrect friend a really convincing argument or suggest an experiment to figure this out?

Solution: They are not the same thing. Friend A prepared particles that are each in a superposition of S_z states, while friend B prepared particles where half are in a pure state $|S_z; +\rangle$ and the other half are in a different pure state $|S_z; -\rangle$. To test if they are the same thing or not, we could pass the beam through a Stern-Gerlach setup in the x-direction. Friend A's beam will yield 100% $|S_x; +\rangle$, while friend B's beam will yield 50% $|S_x; +\rangle$ and 50% $|S_x; -\rangle$. Note that if we pass the original beams through a Stern-Gerlach setup in the z-direction, they would both yield the same result, 50% $|S_z; +\rangle$ and 50% $|S_z; -\rangle$.

- 8.) You have seen that the time-evolution operator $e^{-i\hat{H}t/\hbar}$ is unitary. (a) What does the for of this operator tell you about the time-dependence of the expectation value of energy? Interpret this physically.

Solution: For any arbitrary state $|\psi\rangle$ with initial expectation value of energy $E = \langle\psi|\hat{H}|\psi\rangle$, then the time-evolved expectation value of energy will be $\langle\psi(t)|\hat{H}|\psi(t)\rangle = \langle\psi|e^{i\hat{H}t/\hbar}\hat{H}e^{-i\hat{H}t/\hbar}|\psi\rangle$. Note that $e^{i\hat{H}t/\hbar}$ commutes with $\hat{H}$, since you can expand the exponential into different terms with different powers of $\hat{H}$, and each such term trivially commutes with $\hat{H}$, since $\hat{H}$ commutes with itself. Thus $\langle\psi(t)|\hat{H}|\psi(t)\rangle = \langle\psi|e^{i\hat{H}t/\hbar}\hat{H}e^{-i\hat{H}t/\hbar}|\psi\rangle = \langle\psi|\hat{H}e^{i\hat{H}t/\hbar}e^{-i\hat{H}t/\hbar}|\psi\rangle = \langle\psi|\hat{H}|\psi\rangle = E$

Thus, the expectation value of energy does not change. Of course, it should not change, if energy is conserved!

- 9.) Homework 3, Problem 4: 2-qubit Hamiltonian

For a closed quantum system consisting of two spin-1/2 particles, the Hamiltonian of the system is given by the Heisenberg model

$$\hat{H} = J\vec{S}_1 \cdot \vec{S}_2$$

- Prove that each component of the total spin $S_x^{tot}, S_y^{tot}, S_z^{tot}$ commutes with the Hamiltonian.
- Prove that the total spin S_{tot}^2 commutes with the Hamiltonian
- What does it mean that the Hamiltonian, which is Hermitian, commutes with another Hermitian operator?